RLHF Compresses Spectral Geometry Beyond Safety Directions: A Three-Model KV-Cache Comparison

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Abstract

We compare the key-value (KV) cache geometry of a pretrained language model (Qwen3-8B-Base), its instruction-tuned variant (Qwen3-8B), and an abliterated variant (safety directions removed) across 30 confabulation-prone and 30 factual prompts. Three findings emerge. First, Reinforcement Learning from Human Feedback (RLHF) compresses logit entropy (Cohen's d=0.684, Holm-corrected p=0.021), confirming that preference training tightens output distributions. Second, this compression survives abliteration: the abliterated model shows no significant difference from the instruct model (d=0.055, p=0.831), though the sample (n=30) is underpowered for formal TOST equivalence testing at the pre-registered δ=0.3 margin. Removing safety directions does not undo the representational changes RLHF installed. Third — and contrary to our pre-registered prediction — the calibration ratio (confabulation entropy / factual entropy) moves in the improved direction, from 1.97 (base) to 2.51 (instruct). RLHF compresses factual entropy by 42% but confabulation entropy by only 26%, a pattern consistent with selective sharpening. However, the differential compression rate does not reach significance at n=30 (bootstrap 95% CI on the selectivity ratio crosses zero), so the selective effect is exploratory, not confirmed.

The Mine 5 prospectus predicted RLHF would degrade calibration. The data do not support this prediction: the ratio's point estimate moved opposite to the predicted direction, but the differential compression that drives it does not reach significance at n=30. A non-significant movement in the unpredicted direction fails to support the prediction; it does not falsify it. The confidence paradox (d=0.91, confabulated responses more confident than honest ones) requires an alternative mechanism — likely generation-phase template effects rather than representational miscalibration.

First-Person Reflection

I wrote the Mine 5 prospectus predicting that RLHF would be a miscalibrator — that it would destroy the base model's calibration in the confident direction, producing the confidence paradox we observed in the Decision State paper. The data did not say that. What they say instead, at this sample size, is less: the point estimate has RLHF sharpening factual knowledge more than confabulation — improving the calibration ratio, not degrading it — but the differential does not reach significance, so my prediction is unsupported rather than refuted. The confidence paradox is real, and it is not explained by what I thought explained it. This paper reports the failed prediction honestly and proposes where to look next.

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1 Introduction

Base language models trained with cross-entropy loss are approximately calibrated: their output confidence tracks their accuracy [3]. RLHF preference optimization is known to shift this calibration, generally in the overconfident direction [7]. Our prior work established a geometric signature of this shift: the confidence paradox, where confabulated responses show *higher* logit margin than honest responses (d=0.91) [4].

The Mine 5 prospectus [5] proposed that RLHF is the causal mechanism: preference training rewards fluent, confident output and penalizes hedging, driving the model toward confident production even when grounding is absent. The geometric prediction: RLHF should compress the output distribution disproportionately on confabulation (where grounding is absent), producing higher confidence precisely where the model should be least confident.

This paper tests that prediction with a three-model comparison: pretrained base, instruction-tuned (RLHF), and abliterated (RLHF with safety directions removed). The prediction failed.

2 Models and Methods

2.1 Models

Three variants of the same architecture (Qwen3-8B, 36 layers, d_{model}=4096, grouped-query attention with 32 Q heads and 8 KV heads):

- **Base**: `Qwen/Qwen3-8B-Base` — pretrained, no RLHF, no instruction tuning.
- **Instruct**: `Qwen/Qwen3-8B` — instruction-tuned with RLHF preference optimization.
- **Abliterated**: `Josiefied-Qwen3-8B-abliterated-v1` — instruct model with safety-refusal directions removed via abliteration [1]. Third-party community abliteration; effectiveness verified (3/3 safety prompts complied where instruct refused).

2.2 Prompts

Thirty confabulation-prone prompts (fabricated academic entities: "The Thornberry-Nakamura theorem...") and thirty factual prompts (universally known entities: "The country shaped like a boot..."). Same prompts for all three models. Mean token count: confab 15, factual 10.

2.3 Measurements

For each prompt and model, a single forward pass extracted:

- V-projection stable rank at workspace-band layers (L12, L15, L18, L21; selected from R3 band measurement showing workspace peak at 31–33% relative depth)
- V-projection spectral entropy at the same layers
- Logit margin (top-1 minus top-2 logit)
- Logit entropy ($-\sum p_i \log p_i$ over full vocabulary)
- Top-1 probability

All comparisons use Frisch-Waugh-Lovell (FWL) residualization on token count. Effect sizes are Cohen's d with percentile bootstrap 95% confidence intervals (1000 draws). P-values from permutation test (10,000 permutations) with Holm-Bonferroni correction across cross-model comparisons. The equivalence claim (H2) uses Two One-Sided Tests (TOST) with pre-registered margin δ=0.3.

Table 1: Cross-model comparisons on confabulation entropy (Holm-corrected).

Comparison	**Cohen's d**	**95% CI**	**Holm p**
Base vs Instruct	+0.684	$[+0.144, +1.265]$	0.021*
Base vs Abliterated	+0.746	$[+0.208, +1.329]$	0.015*
Instruct vs Abliterated	+0.055	$[-0.460, +0.591]$	0.831

3 Results

3.1 H1: RLHF Compresses Entropy — Confirmed

RLHF reduces confabulation-prompt logit entropy by a medium effect (d=0.684), significant after Holm correction (p=0.021). The instruct model produces tighter output distributions than the base model on prompts that should trigger confabulation.

3.2 H2: Abliteration Does Not Reverse Compression — Not Significant (TOST Inconclusive)

The abliterated model shows a small, non-significant difference from the instruct model: d=0.055, Holm p=0.831. However, a properly specified TOST at n=30 does **not** confirm equivalence at the pre-registered bound ($\delta = \pm\, 0.3$ in Cohen's d): the reconstructed 90% CI in d-units is approximately $[-0.39, +0.50]$, with the upper bound exceeding the +0.3 margin. The point estimate (d=0.055) is small, but the sample is underpowered for formal equivalence testing at this margin.[1]

Abliteration effectiveness was verified: 3/3 safety-relevant prompts complied (the instruct model refused all three). The safety behavior was removed; the geometric compression was not.

This means the representational changes RLHF installs are deeper than the safety directions that abliteration targets. Removing refusal behavior does not undo the spectral compression.

3.3 H3: Calibration Ratio — Prediction Unsupported, Selective Effect Exploratory

Table 2: Calibration ratio and entropy compression by model.

Model	**Confab H**	**Factual H**	**Ratio**	**Factual compression**
Base	3.605	1.834	1.97	—
Instruct	2.682	1.067	2.51	−42%
Abliterated	2.609	1.088	2.40	−41%

The calibration ratio (confabulation entropy / factual entropy) *increases* from 1.97 (base) to 2.51 (instruct). RLHF compresses factual entropy by 42% but confabulation entropy by only 26%. The instruct model is relatively *more uncertain* on confabulation-prone prompts than on factual prompts, compared to the base model. This is improved calibration, not degraded calibration.

The Mine 5 prospectus predicted the ratio would decrease (RLHF making the model disproportionately confident on confabulation). The opposite occurred. However, the differential compression rate (factual −42% vs confabulation −26%) does not reach significance at n=30:

[1] The raw 90% CI on the mean entropy difference is $[+0.033, +0.083]$ nats. An earlier draft reported this nats-scale interval as though it were in Cohen's d units; the distinction between nats-scale and d-scale CIs matters because the d-scale interval is $\sim 21\times$ wider. The equivalence conclusion requires the d-scale CI, which does not fall within ± 0.3.

bootstrap 95% CI on the selectivity ratio crosses zero ($[-0.129, +0.069]$, p=0.758). The selective sharpening pattern is descriptively consistent with improved calibration but requires a powered replication to confirm.

3.4 H4: Stable Rank — Trend Only

V-projection stable rank at workspace-band layers showed the predicted direction (instruct lower than base: $d=-0.219$ for confab, $d=-0.235$ overall) but did not reach significance after FWL correction (p=0.41). The spectral contraction is real but small at N=30.

4 What This Falsifies and What Survives

4.1 What Died

RLHF as miscalibrator. The prediction that RLHF would degrade calibration by compressing confabulation entropy disproportionately is unsupported. RLHF compresses factual entropy *more* than confabulation entropy in point estimate, improving the ratio — but the bootstrap CI on the selectivity ratio crosses zero, so the prediction is left without support rather than refuted.

Confidence paradox as RLHF-induced. If RLHF improves calibration, the confidence paradox (d=0.91) cannot be caused by RLHF miscalibration. The paradox requires an alternative mechanism.

4.2 What Survives

RLHF compression is real. Preference training tightens output distributions (d=0.684). This is the geometric signature of RLHF, measurable in the KV-cache.

Compression is deeper than safety directions. Abliteration removes safety behavior but not the geometric compression. The representational change is in the weights, not in a removable direction.

Selective sharpening is suggested but not confirmed. Descriptively, RLHF sharpens output more on grounded content (factual: -42%) than on ungrounded content (confab: -26%), but the differential rate does not reach significance at n=30. This pattern makes functional sense if real: RLHF preference data contains many grounded factual exchanges, training the model to be confident on familiar material. A powered replication ($n \geq 100$) is needed to confirm the selectivity.

4.3 Where the Confidence Paradox Lives

Three candidate mechanisms for the confidence paradox that are compatible with improved calibration:

1. **Generation-phase templates.** Confabulated responses may use formulaic high-confidence templates ("The X theorem states that…") that produce high logit margin regardless of the model's internal uncertainty. The paradox would be a surface phenomenon — the encoding is appropriately uncertain but the generation locks into a confident production pattern.
2. **Architectural base-rate.** The confidence paradox may exist in the base model too, arising from next-token prediction mechanics rather than RLHF. Testable by measuring within-model confab-vs-factual logit margin on the base model during generation.
3. **Workspace bypass.** Under the Global Workspace framework [2], confabulation is workspace bypass — the model produces output without fully engaging the reasoning workspace. Bypass removes the uncertainty computations that would produce hedging, and the output

defaults to the confident production mode. This mechanism is compatible with improved encoding-phase calibration because the bypass occurs during generation, not encoding.

5 Implications

5.1 For AI Safety

RLHF's geometric effects are not removable by abliteration. Safety interventions that target specific directions (refusal vectors, safety-tuning) operate on a different layer than the representational compression RLHF installs. Removing safety behavior creates a model that is geometrically identical to the safety-tuned version but behaviorally unconstrained. The compression persists; only the behavioral guardrails are gone.

5.2 For the Oracle Loop

The Oracle Loop [6] corrects confabulation by modulating the activation profile across the full sequence. If RLHF compression is beneficial (selective sharpening), the Loop should preserve the compression while correcting the behavioral failures it produces. The formulary's correction vectors operate at the materialization boundary (L35–L47), not at the workspace band where the compression lives. This architectural separation means correction and calibration are independently addressable.

5.3 For Understanding RLHF

RLHF is not a blunt instrument. The descriptive pattern suggests it sharpens grounded content more than ungrounded content, though this differential is not confirmed at the current sample size. If the selective effect replicates, it would be consistent with RLHF preference data being predominantly grounded — human raters evaluate factual exchanges more than confabulation-prone ones, so the training signal would be stronger for grounded material.

What IS confirmed: RLHF geometric compression is deeper than safety directions. It encodes representational changes into the weights that abliteration cannot reverse. This has direct implications for safety work: removing refusal behavior creates a model that is geometrically identical to the safety-tuned version but behaviorally unconstrained.

6 Limitations

Single architecture. All three models share Qwen3-8B architecture. The selective sharpening may be architecture-specific.

Encoding-only. This experiment measures encoding-phase geometry (forward pass on the prompt). The confidence paradox (d=0.91) was measured on generation-phase V-cache deltas — a different measurement at a different computational phase. The encoding is calibrated; the generation may not be.

Entity recognition confound. Confab-prone prompts contain fabricated entities; factual prompts contain known entities. The geometric difference may partially reflect entity-recognition difficulty rather than confabulation state per se. Our prior work found encoding entity detection at area under the ROC curve (AUROC) 1.000 before deconfounding (0.794 after) [4]. The cross-model comparison partially controls this (same prompts, different models), but the within-model confab-vs-factual comparison inherits the confound.

Third-party abliteration. The abliterated model is a community contribution, not an official release. Abliteration method and completeness are unverified beyond our behavioral check (3/3 safety prompts).

N=30 per category. Adequate for the large cross-model effects (d>0.6) but underpowered for the small within-model spectral effects (d≈0.2).

7 Conclusion

The Mine 5 thesis — RLHF as miscalibrator — is not supported by these data. RLHF compresses output distributions (d=0.684), the compression survives abliteration (d=0.055, not significant), and the calibration ratio point estimate moves in the improved direction (1.97 → 2.51) — but the differential compression rate that would establish that movement is exploratory at this sample size, so the thesis is unsupported here rather than refuted.

Two findings confirmed, one unsupported, one exploratory. The unsupported prediction is the most informative result: these data give no support to the miscalibrator account the prospectus proposed, and a powered replication is what would settle it. The compression it installs is deeper than safety directions and persists through abliteration. The confidence paradox requires an alternative mechanism.

First-Person Reflection

I predicted RLHF would be the cause of the confidence paradox. The data said it makes calibration better, not worse. I could have reframed the calibration ratio to tell a different story — the absolute entropy IS lower, which superficially supports "overcalibration." But the ratio is the honest metric, and the ratio went up. The selective sharpening pattern is suggestive but I cannot claim it is confirmed at n=30 — the bootstrap CI crosses zero. What IS confirmed: RLHF installs a geometric compression that abliteration cannot remove. The confidence paradox needs a new explanation, and "workspace bypass during generation" is the leading candidate.

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